



PFSense Firewall Deployment AND CONFIGURATION

Centralized Network Security using pfSense Firewall

pfBlockerNG Setup, GeoIP Blocking and DNSBL Website Filtering

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Project Title

Deployment and Configuration of pfSense Firewall with pfBlockerNG, GeoIP Blocking and DNSBL Filtering

Objective

As a SOC Intern at ITSimplera Institute my task was to deploy and configure the pfSense firewall and build a secure network environment. I installed pfBlockerNG, set up GeoIP based country blocking, created DNSBL rules to filter websites and configured administrator access for the WebGUI. This task gave me a foundational understanding of firewall administration and network traffic filtering, and it builds the base for the Blue Team activities coming next.

Requirements Covered

- Deploy the pfSense firewall virtual appliance
- Configure pfSense for network security
- Install and configure the pfBlockerNG package
- Configure GeoIP based country blocking
- Configure DNSBL to block specified websites
- Create and validate administrator access rules for the WebGUI
- Monitor firewall activity using pfBlockerNG logs and reports
- Generate and validate firewall events by testing the blocking rules
- Document each step of the process

1. pfSense Deployment

pfSense can be deployed in a few different ways, but for this lab I created a new virtual machine in VirtualBox and installed pfSense using the official Netgate installer ISO. I gave the VM the name pfsense and selected FreeBSD 64 bit as the operating system since pfSense is built on FreeBSD.

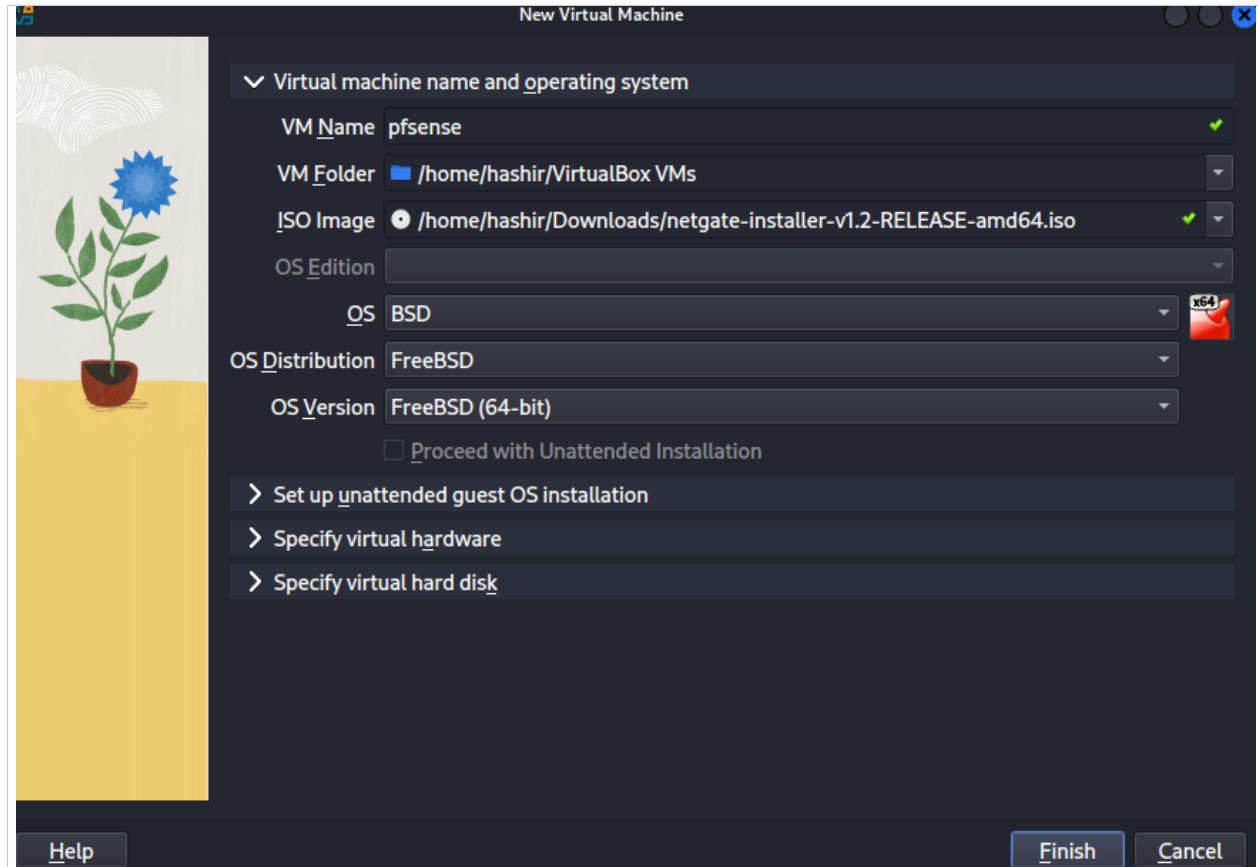


Figure 1: New virtual machine created in VirtualBox with the Netgate installer ISO attached

Before starting the install I opened the network settings of the VM so it could reach the internet and also talk to the other machines on the internal network. Adapter 1 was set to NAT for internet access, this lets the firewall reach the pfBlockerNG update servers later.

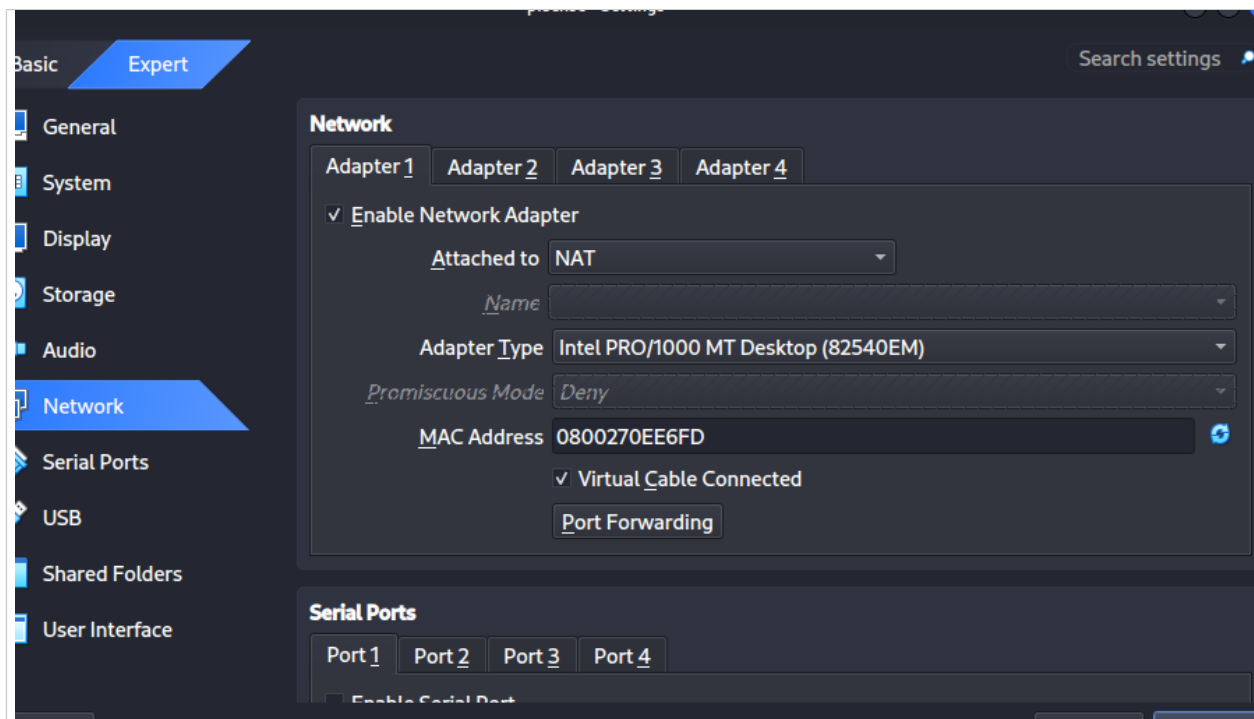


Figure 2: Network adapter 1 configured as NAT with the default Intel PRO/1000 adapter

The virtual hard disk was attached as an IDE primary device with 20 GB of storage, which is enough for a lab firewall build.

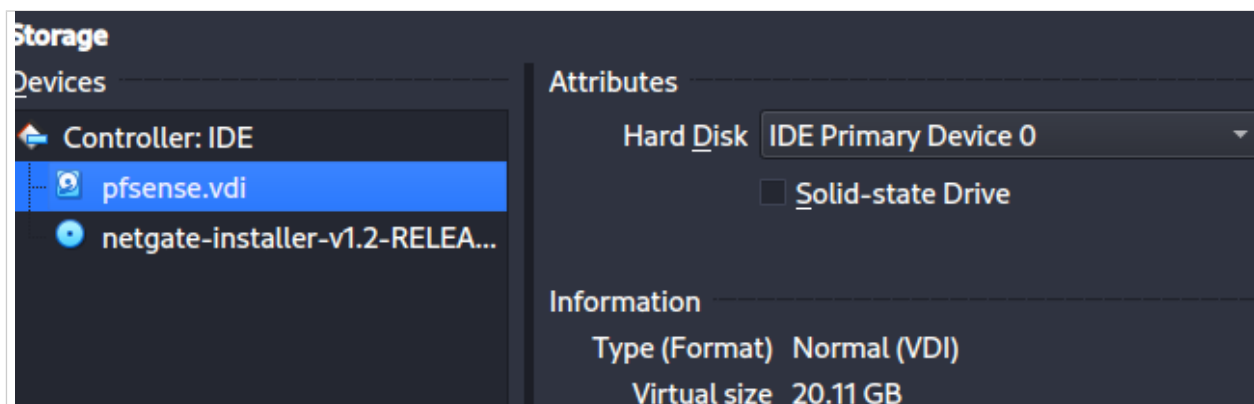


Figure 3: Storage settings showing the pfsense virtual disk and the installer ISO

During the install the Netgate installer asked which disk to use for the software. I selected the 20 GB VBOX HARDDISK and kept the default UFS file system and GPT partition scheme.

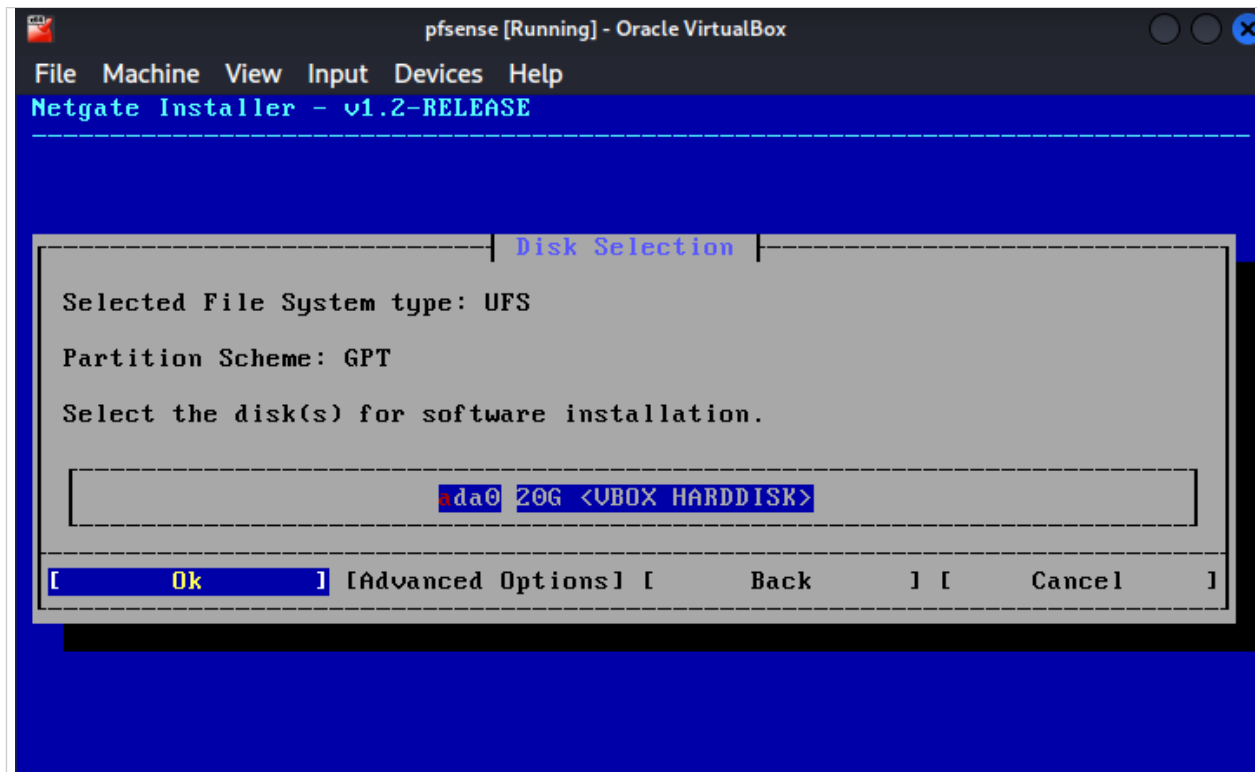


Figure 4: Disk selection step of the Netgate installer

After the install finished and the machine rebooted, pfSense 2.8.1 RELEASE booted up and automatically picked up an address on the WAN interface through DHCP, and the LAN interface was assigned the default 192.168.1.1 network. The console menu also confirmed the bootup was complete.

```
pfSense [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Starting CRON... done.
pfSense 2.8.1-RELEASE amd64 20260701-1525
Bootup complete

FreeBSD/amd64 (pfSense.home.arp) (ttyv0)
VirtualBox Virtual Machine - Netgate Device ID: 8adcd710fb677b2642ff

*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

WAN (wan) -> em0 -> v4/DHCP4: 10.0.2.15/24
v6/DHCP6: fd17:625c:f037:2:a00:27ff:fea4:9747/64
LAN (lan) -> em1 -> v4: 192.168.1.1/24

0) Logout / Disconnect SSH          9) pfTop
1) Assign Interfaces                10) Filter Logs
2) Set interface(s) IP address      11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults        13) Update from console
5) Reboot system                   14) Enable Secure Shell (sshd)
6) Halt system                     15) Restore recent configuration
7) Ping host                       16) Restart PHP-FPM
8) Shell

Enter an option: █
```

Figure 5: pfSense console after boot showing the WAN and LAN interface addresses

2. Initial Configuration

Once pfSense was up I opened the LAN address in the browser to reach the WebGUI. The setup wizard walked me through the general settings for the firewall, including hostname, domain and the DNS servers used by the firewall itself.

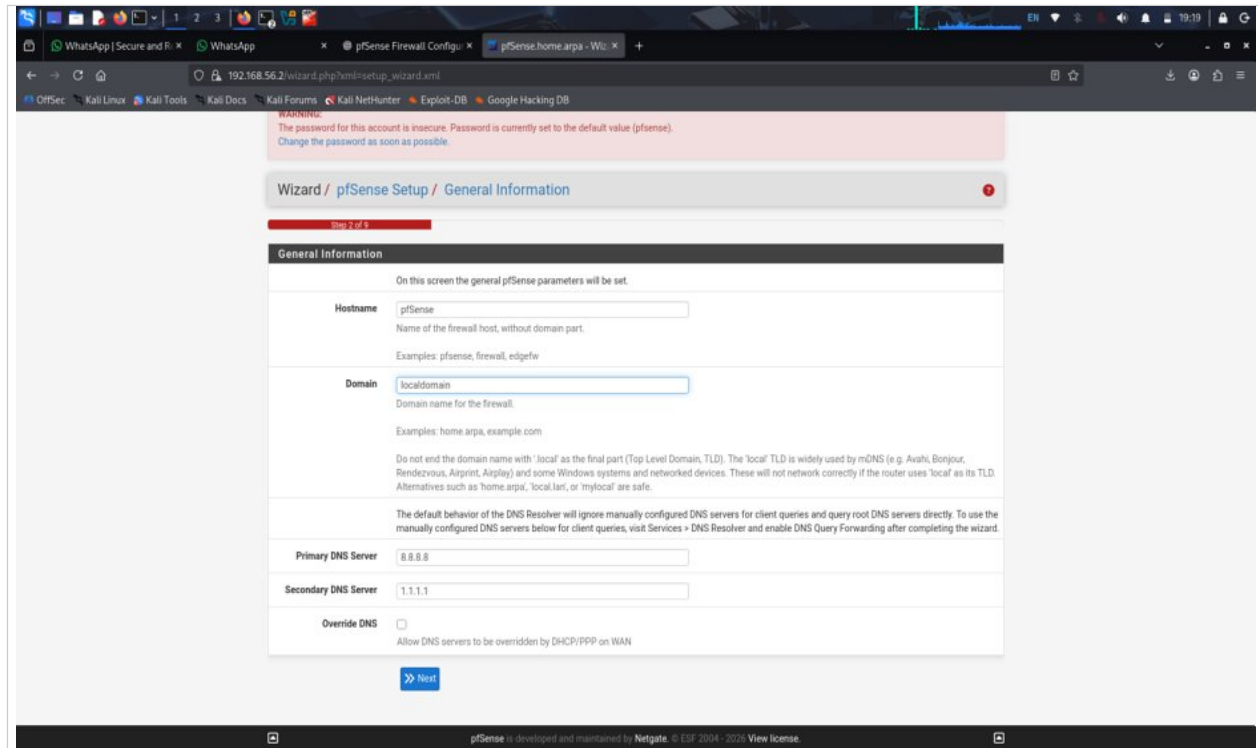


Figure 6: General Information step of the pfSense setup wizard

After finishing the wizard the dashboard loads and shows the status of both interfaces. The WAN interface got an address from DHCP and the LAN interface is set to 192.168.56.2 in this lab, which is the address used to reach the WebGUI for the rest of the configuration.

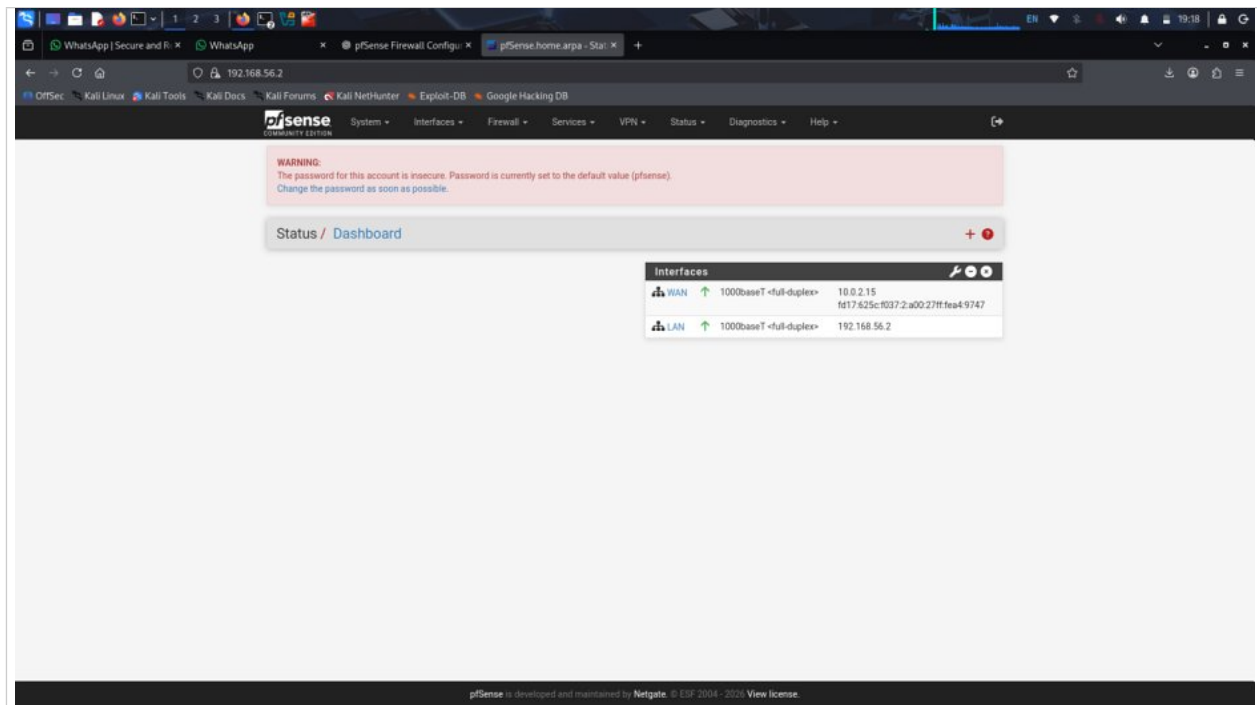


Figure 7: Dashboard showing the WAN and LAN interfaces after the wizard was completed

I also noticed the warning banner at the top of the dashboard telling me the admin password is still set to the default value. This is addressed later in the administrator access control section of this report.

3. Installation of pfBlockerNG

pfBlockerNG is not installed by default so it has to be added through the Package Manager. I went to System, Package Manager, Available Packages and searched for pfblocker. Two options came up, the stable pfBlockerNG package and the pfBlockerNG devel package, I installed the regular pfBlockerNG package for this lab since it is the more stable release.

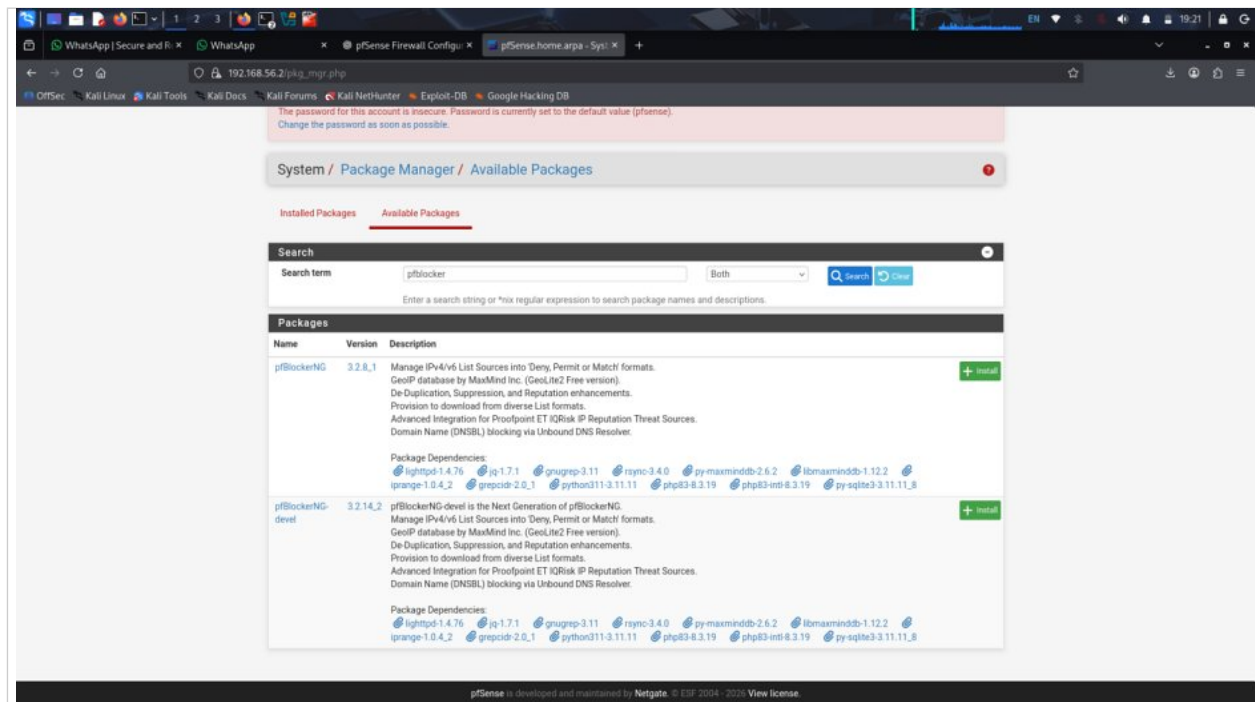


Figure 8: Available Packages search results for pfblocker in the Package Manager

The package pulled in a number of dependencies automatically, such as php, sqlite and the GeoIP database libraries. Once the install finished a new pfBlockerNG menu appeared under Firewall in the WebGUI.

4. GeoIP Based Country Blocking

With pfBlockerNG installed the next step was to turn on GeoIP filtering. Under Firewall, pfBlockerNG, GeoIP I enabled the GeoIP module and selected the countries that should be blocked, for this lab I added the Netherlands as an example country along with a couple of other high risk regions.

- Enabled the GeoIP module and set it to Deny Inbound and Outbound
- Added the target country alias, for example NL for Netherlands
- Saved the settings and let pfBlockerNG build the alias and generate the firewall rule automatically

Once saved, pfBlockerNG creates the alias table for the selected country and adds a firewall rule on the WAN interface that drops traffic coming from that country. The rule can be seen afterwards under Firewall, Rules, WAN, and it updates automatically on the schedule set in the cron settings.

5. DNSBL Website Filtering

DNSBL, or DNS Black List, is the pfBlockerNG feature used to block access to specific websites by domain name. To use it, pfBlockerNG needs a Virtual IP configured on the loopback interface, this is the address the DNS resolver redirects blocked domains to. I created this VIP under Firewall, Virtual IPs using the IP Alias type on the Localhost interface with the address 127.0.0.1.

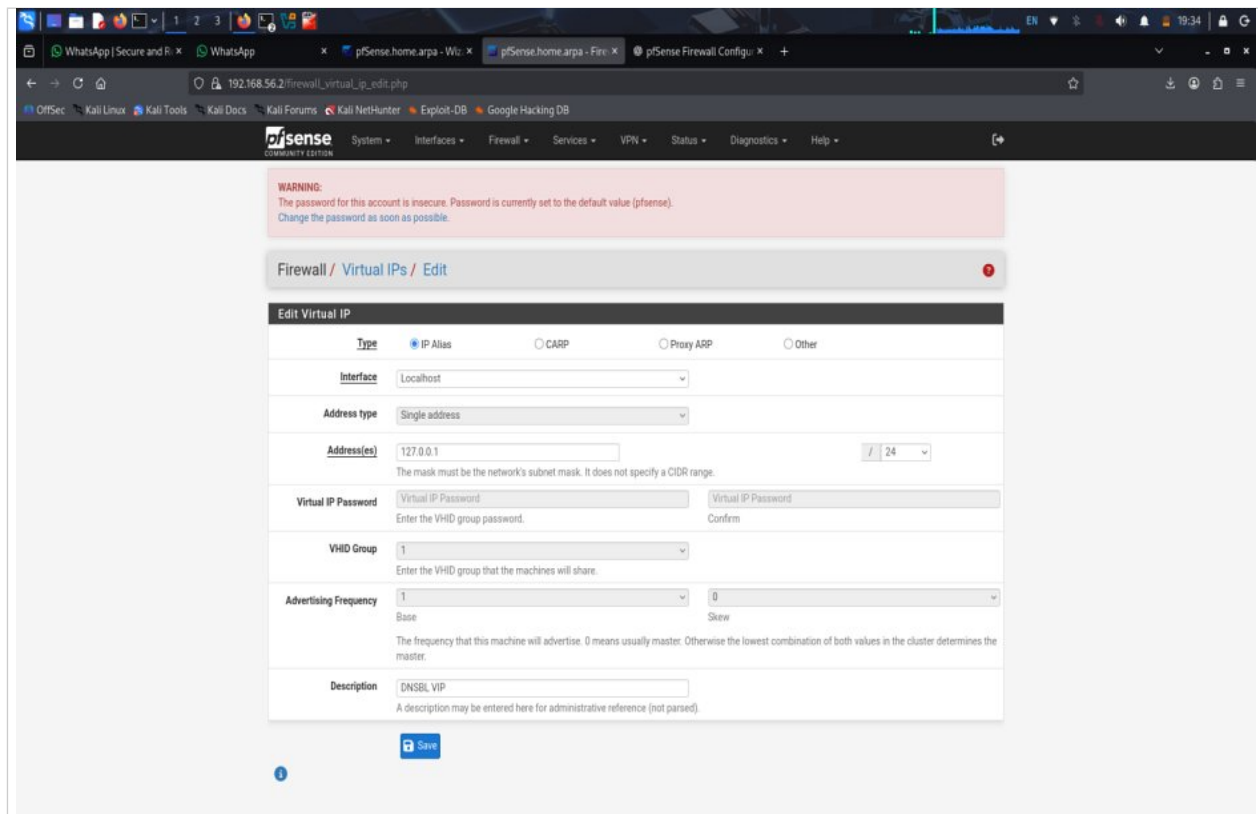


Figure 9: DNSBL virtual IP configured as an IP Alias on the Localhost interface

After saving, the Virtual IPs page listed the new 127.0.0.1 alias with the description DNSBL VIP, and pfSense showed a message that the change needs to be applied before it takes effect.

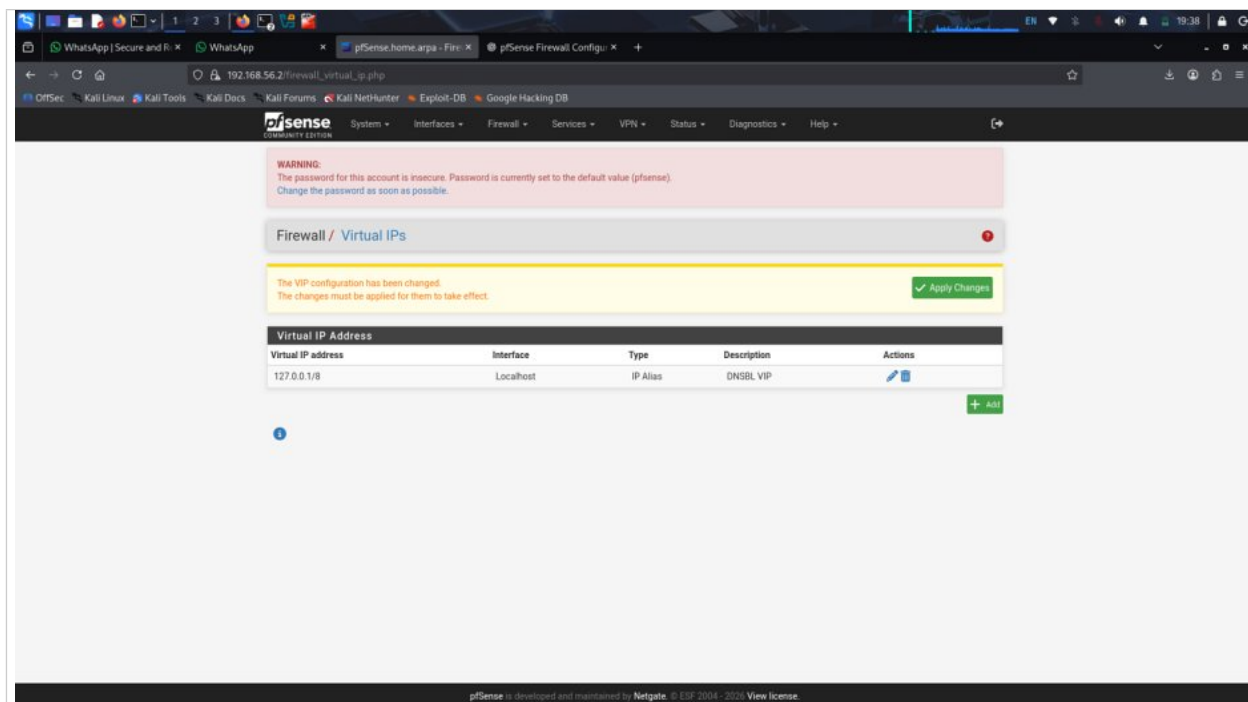


Figure 10: Virtual IP list showing the DNSBL VIP after it was applied

With the VIP in place I went to the DNSBL tab under pfBlockerNG and enabled it, then added the domains that needed to be blocked to a Custom List, in this lab facebook.com and chat.com were added as the test entries. Once the DNSBL group was force updated, the domains resolve to the loopback address instead of their real IP, which stops the site from loading on any client behind the firewall.

6. Administrator Access Control

Securing the WebGUI itself is just as important as the traffic rules. The default install still uses the default admin password, this warning shows up on every page until it is changed, so the first step was going to System, User Manager and setting a strong password for the admin account.

- Changed the default admin password under System, User Manager
- Restricted WebGUI access to HTTPS only under System, Advanced, Admin Access
- Limited WebGUI access to the LAN interface so it cannot be reached from WAN
- Enabled login protection so repeated failed logins get temporarily blocked

These changes make sure the firewall itself cannot be taken over from the internet side, and that only an admin on the internal network with the correct password can reach the management interface.

7. Monitoring pfBlockerNG Logs

pfBlockerNG keeps a log of every update cycle, this is useful to confirm the block lists were downloaded correctly and that the DNSBL and GeoIP rules are active. I forced an update run from Firewall, pfBlockerNG, Update and reviewed the log output.

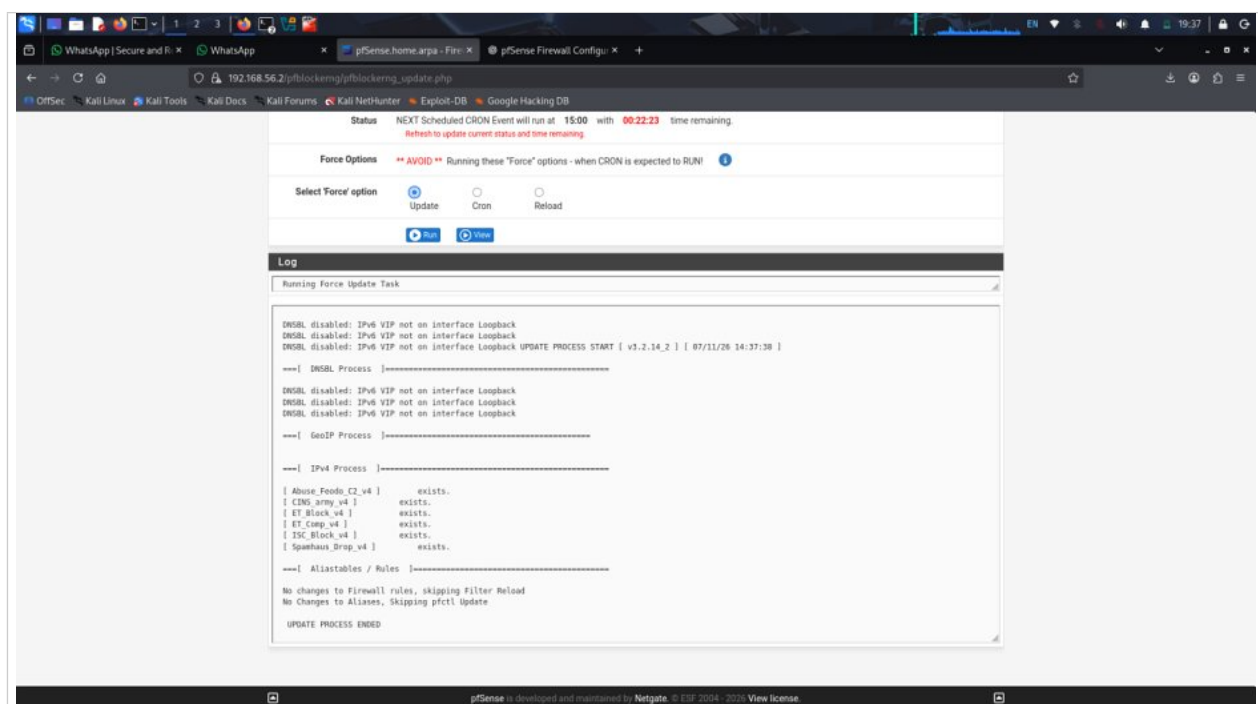


Figure 11: pfBlockerNG force update log showing the DNSBL, GeoIP and IPv4 alias processing

The log confirms the IPv4 threat lists such as ET_Block, CINS_army and Spamhaus_Drop already exist and did not need to be redownloaded, and it also shows the GeoIP and DNSBL sections running without errors, other than a note that the DNSBL IPv6 VIP is not bound to the loopback interface, which is expected since this lab only uses IPv4.

8. Generating and Validating Security Events

To confirm the rules actually work I tested both the country block and the website block from a client machine behind the firewall.

- Tried browsing to facebook.com and chat.com from a LAN client, both requests failed to resolve since DNSBL pointed them to the loopback address
- Used a VPN endpoint in the blocked country to confirm the connection was dropped by the GeoIP rule on WAN
- Checked Status, System Logs, Firewall to see the blocked connection entries, and Firewall, pfBlockerNG, Alerts to see the matching DNSBL and GeoIP hits

Both tests confirmed the traffic was stopped as expected and the matching alert showed up on the pfBlockerNG dashboard, which means the whole pipeline works end to end, from the block list, to the firewall rule, to the logged alert.

9. Conclusion

This task walked me through a complete small scale firewall deployment, from installing pfSense in a virtual lab to installing pfBlockerNG and setting up GeoIP and DNSBL based filtering. I also secured the WebGUI itself and confirmed the blocking rules generate real alerts that can be seen and analyzed on the dashboard. It gave me a much better hands on understanding of how network traffic filtering and firewall administration works in practice, and this is a good base to build on for the next Blue Team and network security tasks.